WILDFIRE PREVENTION · CBMGO

Does a 3-year fire rule hold in the Cerrado?

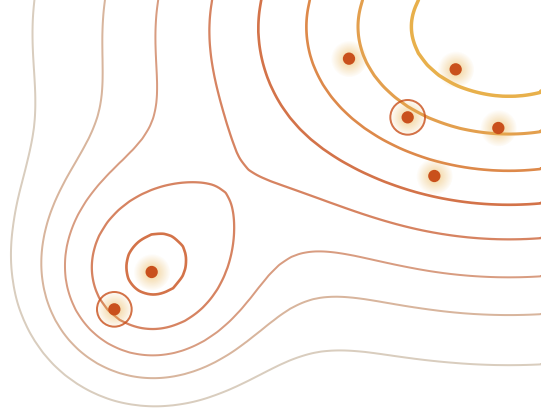
Testing a field assumption against ten years of satellite heat hotspots in Goiás state parks

22.7% of burned areas burned again within 3 years



Executive presentation | INPE heat hotspots 2016–2025 | 2,771 hotspots | 9 state parks | Heat hotspot | Re-burn | ILLUSTRATIVE

Fire protects an area for about one year, not three



CBMGO weighed dropping burned areas from inspection rounds for three years, on the assumption that fuel needs that long to recover.

PROTECTION AFTER FIRE

6.4% → 12.7%

Annual chance of a new fire doubles from year 1 to year 2 after a fire.

Year 1	6.4%
Year 2	**12.7%**

TIME TO RE-BURN

22.7%

More than 1 in 5 burned areas burned again within 3 years — inside the window the rule would leave unchecked.

Within 2 yrs	12.5%
Within 3 yrs	**22.7%**

WHERE AND WHEN

71% / 91%

Fire concentrates in 2 of 9 parks (Terra Ronca, Serra Dourada) and in August–October.

2 of 9 parks	**71%**
Aug–Oct	**91%**

RECOMMENDATION

Do not adopt the 3-year skip. Keep burned areas on the inspection list from the second year on and prioritize sectors by fire history, season and drought.

NEXT STEP

Pilot a risk-based ranking of park sectors for one fire season and compare it with where new hotspots occur.

Heat hotspots are an indirect fire indicator. Results show association, not causation.

02 | CONTEXT

Can recently burned areas leave the inspection rounds?

01 · LIMITED STAFF

93%

of surveyed commanders cite lack of personnel; CBMGO runs inspection rounds in 9 state parks with limited staff.

02 · FIELD ASSUMPTION

~3 years

assumed for Cerrado fuel to recover after a fire, so burned areas could be skipped in the rounds.

03 · THE CENTRAL QUESTION

Does that three-year interval hold in Goiás state parks, and how should prevention be prioritized instead?

2,771
heat hotspots

9
state parks

10 years
analyzed, 2016–2025

Source: INPE BDQueimadas, NPP-375/VIIRS satellite · 9 Goiás state parks · 2016–2025.

Ten years of satellite data to test a field assumption

SATELLITE HOTSPOTS

2,771

detected inside the 9 parks, 2016–2025

BURNED AREAS

723

~1 km grid cells that burned

RECURRENCES

499

times the same area burned again

OBJECTIVE

Test whether burned areas stay safe for three years, and define where and when to focus prevention.

APPROACH

1. **Structure hypotheses**
2. **Test them with satellite data and statistics**
3. **Turn findings into recommendations**

SCOPE & LIMITATION Satellite hotspots are an indirect indicator; no field data on vegetation was available.

Hotspots in the same 0.01° (~1.1 km) cell less than 180 days apart count as one fire event. Source: INPE BDQueimadas · NPP-375/VIIRS · 2016–2025.

Three conditions must all hold for the 3-year rule to work

HYPOTHESIS UNDER TEST

Burned areas stay safe from new fires for about three years while fuel recovers.

AND

C1 · DATA

The data show 3 years of protection

- Few areas re-burn within 3 years
- Risk stays low in years 2 and 3
- Most common interval is 3+ years
- The pattern holds in every park

EVIDENCE · SATELLITE DATA

C2 · FUEL

Fuel takes 3 years to recover

- Cerrado grass fuel needs about 3 years to come back

EVIDENCE · LITERATURE

C3 · MEASUREMENT

Hotspots measure re-burns

- A recurrent hotspot means the same area burned again

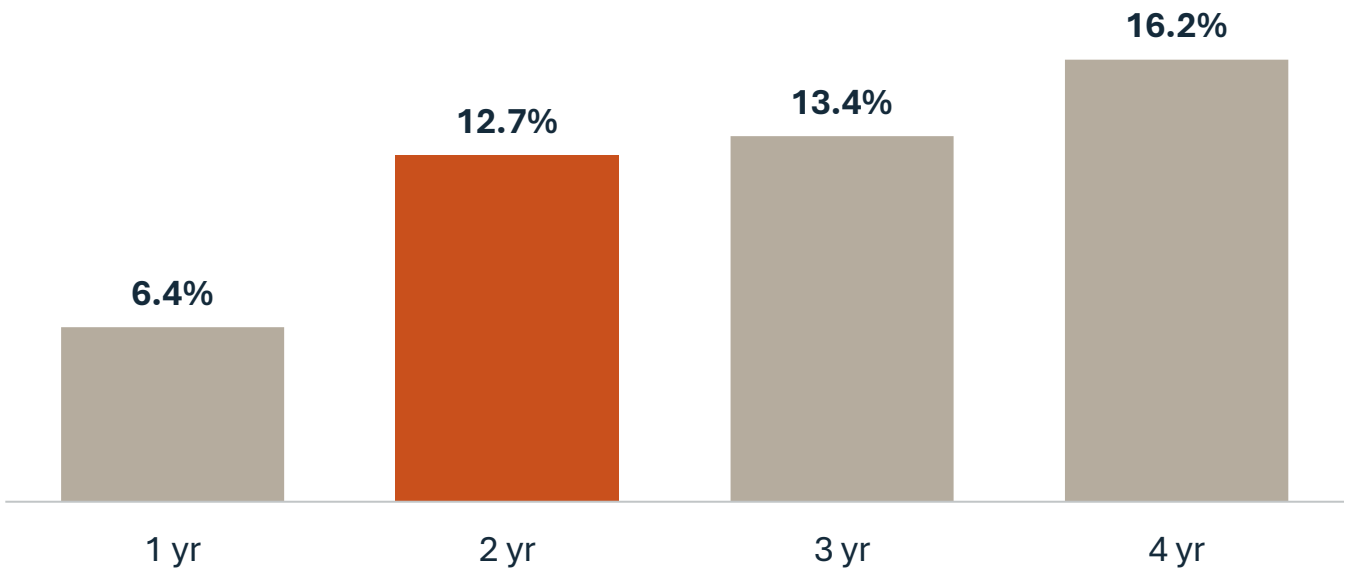
EVIDENCE · SATELLITE DATA

LOGIC One false condition rejects the hypothesis. Evidence: satellite data (C1, C3) and literature (C2).

The risk of a new fire doubles in the second year

Chance of a new fire that year, by years since the last fire

Goiás state parks · 2016–2025



OR = 0.54

Burning one year halves the odds of burning the next ($p < 0.001$).

2× the risk

The chance doubles from year 1 to year 2.

No safe window

No 3-year safe period appears in the data.

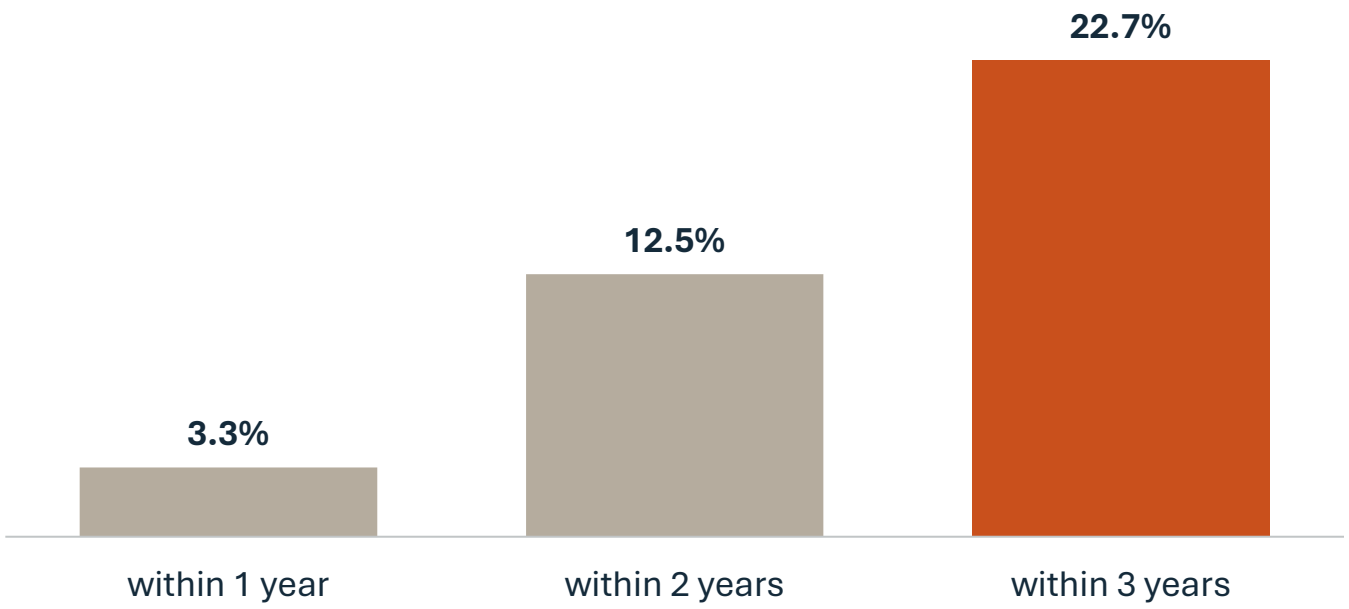
CONCLUSION **Fire buys about one year of protection, not three.**

Logistic regression on 4,937 cell-years, cluster-robust errors by cell. Association, not causation.

More than 1 in 5 burned areas burns again within 3 years

Share of burned areas that burned again, by time since the fire

Kaplan-Meier · 723 areas · 2016–2025



22.7%

burned again inside the 3-year window (95% CI 19.6–25.8%).

12.5%

burned again within only 2 years.

723 areas

followed, including those that never burned again.

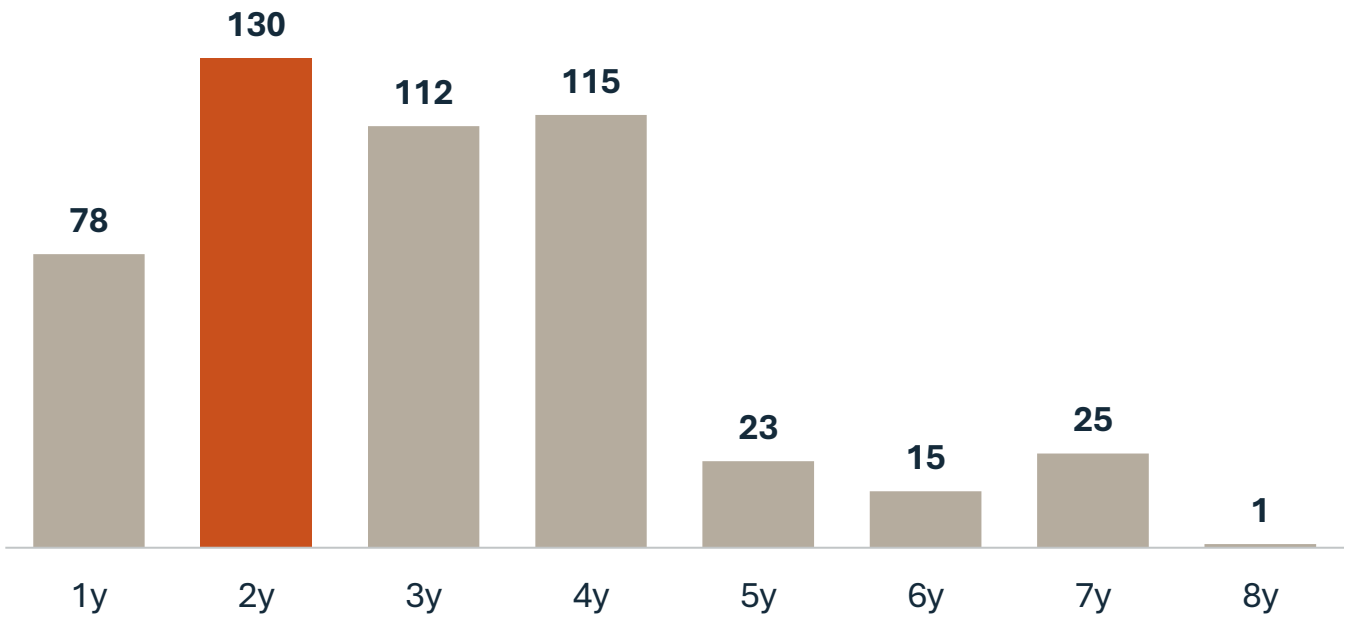
CONCLUSION **A 3-year skip would leave these areas off the inspection list.**

Kaplan-Meier survival analysis; areas not burned again by December 2025 treated as censored.

Three years is the typical gap, not a safe period

Recurrences by interval between fires, in whole years

Count of recurrences · n = 499



Mode = 2 years

The most common interval between fires.

Median ≈ 3 years

This is why the assumption seemed right (Wilcoxon p = 0.26).

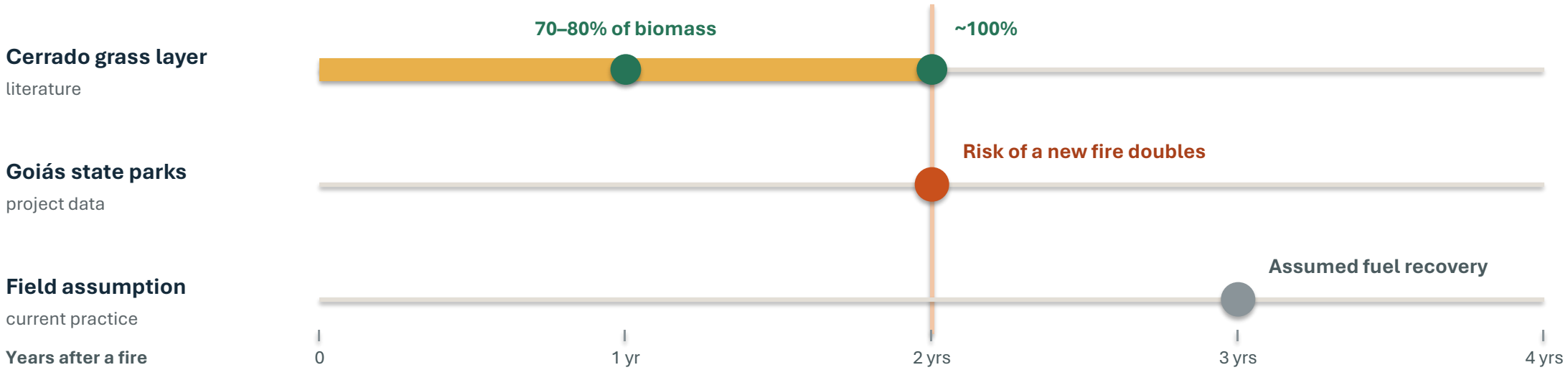
Average ≠ minimum

Protection ends long before the typical gap.

CONCLUSION **The 3 years describe how often fire returns, not how long an area is safe.**

Intervals measured in whole years between consecutive fires in the same ~1 km cell. Cumulative share reaches 64% by year 3.

Grasses explain why protection ends so early



~2 years

for the native grass layer to fully recover after a fire.

Up to 2× taller flames

Invasive grasses (capim-gordura, braquiária) add biomass to the fuel load.

Literature only

This explanation comes from published studies, not from project data.

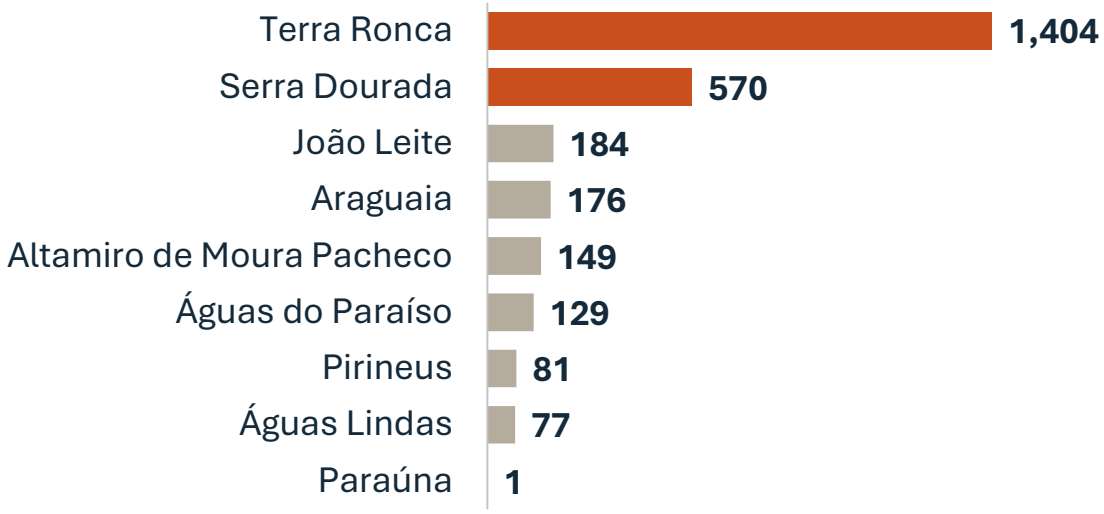
CONCLUSION **The next step is to test grass cover with land-cover data.**

Sources: Gorgone-Barbosa et al. (2016); Onigemo, Miranda & Haridasan (2003); Pivello et al. (2021).

Fire is predictable in space and time

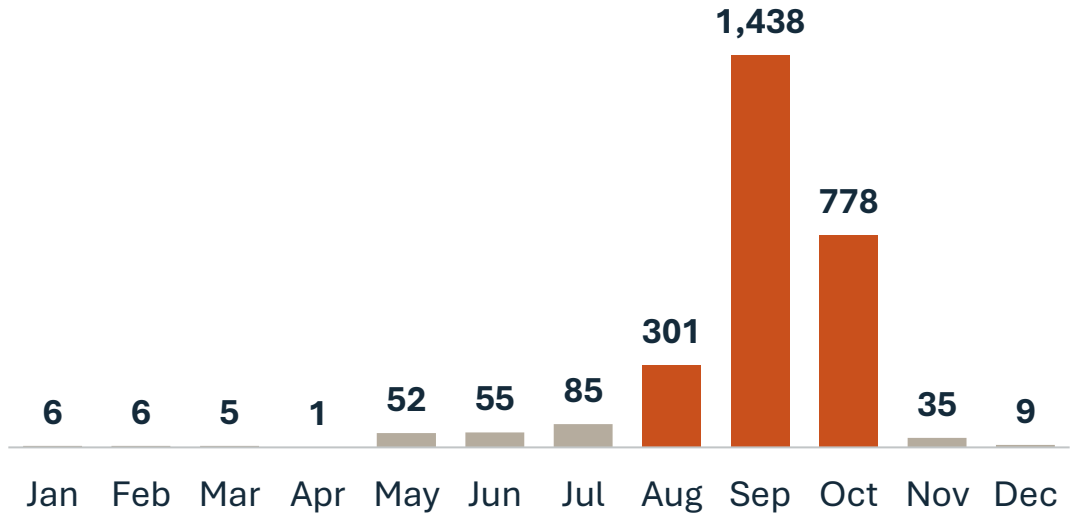
Heat hotspots by park

count · 2016–2025



Heat hotspots by month

count · all years combined



71.2%

of hotspots in 2 of 9 parks

90.8%

between August and October

ρ = 0.83

days without rain vs. hotspots

CONCLUSION **Concentrate prevention in the right parks, before August.**

Source: INPE BDQueimadas, NPP-375/VIIRS · 2,771 hotspots in 9 Goiás state parks. ρ = Spearman correlation, monthly.

Not every finding is relevant enough to guide action

PRIORITIZE FOR ACTION

- Protection lasts ~1 year: the risk doubles in year 2
- 22.7% of burned areas re-burn within 3 years
- 2 parks concentrate 71% of hotspots
- 91% of hotspots fall in August–October
- Drought is strongly associated with hotspots
- Araguaia re-burns faster (median 2.1 years) than Serra Dourada (4.0)

NO SPECIFIC ACTION FOR NOW

- Fire intensity barely differs between frequent and infrequent fire (8.5 vs 9.1 MW)
- Precipitation is not significantly associated with hotspots
- The 2016–2025 hotspot trend is not significant

DECISION **Replace the time-based rule with a risk-based ranking.**

Keep burned areas on the list and prioritize by risk

1

PRIORITY 1
Decide now

Drop the 3-year skip

Keep burned areas on the inspection list from year 2. Decide before the next fire season.

2

PRIORITY 2
Pilot next season

Risk-based sector ranking

Rank sectors by fire history, month and days without rain. Start with Terra Ronca and Serra Dourada.

3

PRIORITY 3
June–July

Pre-season prevention

Firebreaks and patrol schedules in June–July, ahead of the August–October peak.

INVESTIGATION TRACK Test the grass hypothesis with MapBiomas land-cover and burned-area maps.

RECOMMENDATION **Start by dropping the 3-year skip; pilot the ranking in parallel.**

Validate the ranking over one fire season

01

Validate

By June

Rank park sectors using hotspot history, season and drought.

02

Test

August–October season

Apply the ranking in the fire season, keeping burned areas in the rounds.

03

Decide

After the season

Compare the ranking with where new hotspots fell (hit rate and miss rate); keep, adjust or revert.

EXPECTED DECISION

Adopt the risk-based ranking if it captures more new hotspots than the 3-year rule.

Hit rate: share of new hotspots inside top-ranked sectors. Miss rate: share outside them.

CLOSING

Turning satellite data into safer inspection rounds

~1 year

of protection after a fire, not three

22.7%

of burned areas burned again within 3 years

71% | 91%

of hotspots in 2 parks | in August–October

EXECUTIVE RECOMMENDATION

Do not adopt the 3-year skip. Prioritize park sectors by risk and validate the ranking over one fire season.

Fire returns faster than assumed; prevention should follow risk, not the calendar.

CBMGO · INPE heat hotspots 2016–2025 · 2,771 hotspots · 9 state parks

APPENDIX

Supporting analysis and data

A1 Methods: why each statistical test was used

A2 Limitations

A3 Data sources and tools (GitHub · Excel · Power BI)

APPENDIX | A1

Methods: why each statistical test was used

Question	Test	Result
Are the data normal?	Shapiro-Wilk, Kolmogorov-Smirnov (Lilliefors)	No ($p < 0.001$) → non-parametric tests
Is the median interval 3 years?	Wilcoxon	Median 2.98, $p = 0.26$
What share re-burns in 1–3 years?	Kaplan-Meier	3.3% / 12.5% / 22.7%
Does fire lower next-year risk?	Logistic regression	OR 0.54, $p < 0.001$
Is drought associated with hotspots?	Spearman	$\rho = 0.83$, $p < 0.001$
Do parks differ?	Kruskal-Wallis + Dunn (Holm)	$p < 0.001$
Is there seasonality?	Chi-square	90.8% Aug–Oct, $p < 0.001$
Is there a trend?	Mann-Kendall + Sen	Fire events −9/year, $p = 0.049$
Does fire cluster in space?	Moran's I	$I = 0.33$, $p = 0.001$

Significance level $\alpha = 0.05$. Non-parametric tests used because the data are not normally distributed.

APPENDIX | A2

Limitations

Indirect indicator

Heat hotspots are an indirect fire indicator; small or short fires may be missed.

Spatial dependence

Neighboring cells hit by the same large fire are not independent.

No vegetation data

No local vegetation data: the grass explanation relies on the literature.

Missing context

No data on ignition causes; hotspots not normalized by park area.

APPENDIX | A3

Data sources and tools

GitHub

Analysis code and statistical tests

Methods and tests: see A1

Excel

Hotspot base and data treatment

Records: 2,771 hotspots

Power BI

Commanders' fire-risk dashboard

Audience: CBMGO commanders